MIL-B-50898(MU)
7 June 1971

MILITARY SPECIFICATION

BATTERY ASSEMBLY, TRAINING: 16XPM-05

This specification is mandatory for use by all activities of the Army Munitions Command.

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of battery assembly, training: 16XPM-05 used on the XM-15 Trainer for the Atomic Demolition Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-8635 - Decals, for Use on Interior Surfaces
MIL-S-22473 - Sealing, Locking and Retaining Compounds, Single Component

STANDARDS

MILITARY

AN340C5 - Nut
AN515C5 - Screw

FSC - 1125

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MS 29561-010 - "O" Ring
MS 35275 - 247 - Screw

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8887550 - Reservoir Assembly
8887552 - Bottom, Reservoir
8887553 - End, Reservoir
8887554 - Side, Reservoir
8887557 - Vent Plate Assy
8887558 - Block, Orifice
8887559 - Plate, Vent
8887560 - Vent
8887561 - Block
8887562 - 16 X PM-05 Battery Assembly, Training
8887581 - Decal, Battery, Training
8887599 - Screw, Nylon Drain
9206901 - Box, Fiber, For 16 X PM-05 Battery Assembly, Training
9242431 - Cement, Acrylic
9242535 - Sponge Rubber Plug
9242536 - Pad Assembly
ET-8887562 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The battery assembly shall comply with all requirements specified on the drawing (dwg) 8887562, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

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3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristic (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Nut (see AN340C5); Screw (see AN515C5); "O" ring (see MS 29561-010); Screw (see MS35275-247); sealant (see MIL-S-22473); cement, acrylic (see dwg 9242431); Sponge rubber plug see dwg 9242535); screw (see dwg 8887599); pad assembly (see dwg 9242536) covering details of dwgs 8887557 and 8887562 as applicable.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for part in accordance with specification, standard or drawing	Visual
Minor: None defined.		

4.3.2.1.2 Bottom reservoir (see dwg 8887552-1); end reservoir (see dwg 8887553-2); and side reservoir see dwg 8887554) covering details of dwg 8887550.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Length	*Measure
102.	Width	*Measure
103.	Thickness	*Measure
104.	Material (see 3.1)	Visual

*Measure with suitable scale

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Categories	Characteristics	Method of Inspection
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.3 Reservoir assembly (see dwg 8887550) covering a detail of dwg 8887562.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Joints flush	Visual/Manual
102.	Pitch diameter of thread side hole, max	Gage
103.	Minor diameter of thread side hole, max	Gage
104.	Location of side hole	Gage
105.	No excess cement	Visual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.4 Plate, vent (see dwg 8887559) covering a detail of dwg 887557.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Location of large holes	Gage
102.	Location of 2 small holes	Gage
103.	Length, max	Gage
104.	Width, max	Gage
105.	Thickness, max	Gage
106.	Material (see 3.1)	Visual/Manual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.5 Vent (see dwg 8887560) covering a detail of dwg 8887557.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Diameter of left cavity	Gage

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Categories	Characteristics	Method of Inspection
102.	Diameter of right cavity	Gage
103.	Depth of left cavity	Gage
104.	Length from bottom of right cavity to left face	Gage
105.	Diameter of small hole	Gage
106.	Material (see 3.1)	Visual/Manual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.6 Block (see dwg 8887561-1, 2 and 3) covering a detail of dwg 8887557.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Thickness	Gage
102.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.7 Plate, Vent Assembly (see dwg 8887557) covering a detail of dwg 8887562.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Joints flush	Visual/Manual
102.	No loose parts	Visual
103.	Positive terminal red	Visual
104.	No part missing	Visual
105.	Location of center block	Visual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.8 Block, orifice (see dwg 8887558) covering a detail of dwg 8887562.

Categories	Characteristics	Method of Inspection
Critical: None defined.		

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Categories	Characteristics	Method of Inspection
Major:		
101.	Location of counterbored holes	Gage
102.	Diameter of counterbore, max	Gage
103.	Depth of counterbore	Gage
104.	Length, max	Gage
105.	Width, max	Gage
106.	Thickness, max	Gage
107.	Diameter of counterbored holes	Gage
108.	Diameter of countersink on counterbored holes	Gage
109.	Pitch diameter of 4 threaded holes, max	Gage
110.	Minor diameter of 4 threaded holes, max	Gage
111.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.9 Decal, battery, training (see dwg 8887581) covering a detail of dwg 8887562.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for decal in accordance with MIL-D-8635	Visual
Minor: None defined.		

4.3.2.1.10 16XPM-05 battery assembly, training (see dwg 8887562).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	No parts loose or missing	Visual/Manual
102.	Joints flush	Visual/Manual
103.	Reservoir screws below flush	Manual
104.	Decal coated	Visual
105.	No excess cement	Visual
Minor:		
201.	Workmanship (see 3.3)	Visual/Manual

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4.3.2.1.11 Box, fiber, for 16XPM-05 battery assembly, training (see dwg 9206901).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Contents tight in container	Manual
102.	Carton properly sealed	Visual
103.	Marking	Visual
104.	Label properly applied	Visual
Minor: None defined.		

4.3.2.1.11 Packing and Marking for Storage and Shipments (see dwg 8808362) and section 5).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Gross load weight, max	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.1)	Visual

*Determine weight with a suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see SQAP-Appendix I)	Visual/Manual

4.3.3 Inspection testing. The contractor shall perform all of the tests specified in this paragraph.

Test	Test Classification
A. Leakage	Major

This test is not considered destructive and samples so tested shall be returned to the lot.

4.3.3.1 Leakage test (see dwg 8887562). Each battery in the lot shall be subjected to this test (see 4.4.1). If any battery fails to comply with drawing requirement it shall be classified defective and removed from the lot.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8887562. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Leakage (see 4.3.3.1). Fill each cell of the battery with 2.5 cubic centimeters (cc) of water on a level surface. If leakage occurs at the joint between the reservoir and orifice block, dry the battery without disturbing position. Allow the battery to stand 15 minutes, minimum. Observe the battery for any evidence of leakage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The battery shall be packaged in accordance with dwgs 9206901 and 8808362 using the specified moisture barrier.

5.1.2 Level C. The battery shall be packaged in accordance with dwgs 9206901 and 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The battery shall be packed in accordance with dwg 9206901 and 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 9206901 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing Activity
Army - MU

Project No. 1125-A072

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-50898(MU) Battery Assembly, Training: 16XPM-05

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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